NGUYEN VAN HAI

SOFTWARE ENGINEER

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LeetCode Profile
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About me

My name is Nguyen Van Hai. I graduated with a degree in Information Technology from Ho Chi Minh City University of Science - HCMUS, very passionate about technology and system architecture. I am a quick learner with strong problem-solving skills who thrive in a collaborative environment to achieve common goals. I explore new technology and do personal coding projects to perfect my skills. I want to apply my knowledge to new challenges and look forward to learning and develop in a creative working environment.

Education

•VNUHCM - University of Science *2020 - 2024*

Bachelor in Information Technology, Faculty of Information Technology GPA: 3.4

Skills

•Technical Skills

(Technologies, Programming Languages, and Frameworks)

- **Programming Languages**:
  - **JavaScript/TypeScript**: Experience in ExpressJS, NestJS, PassportJS, Prisma, Sequelize, gRPC, and GraphQL. Frontend experience includes ReactJS and NextJS, with authentication using NextAuth.
  - **PHP**: Experience in Moodle development, PHP Unit, and Behat testing.
  - **CSharp / .NET**: Azure Functions (Azure Functions Core Tools) and Entity Framework.
  - **Golang**: Experienced with RESTful APIs using Gin-Gonic, GORM, SQLX, gRPC, and Firebase integration.
- **Algorithms and Data Structures**: Strong knowledge of fundamental algorithms and data structures, essential for efficient application development and optimization.
- **Tools and Technologies**:
  - **Version Control, DevOps**: Git, CI/CD pipelines using Azure DevOps and GitHub Actions.
  - **Containerization, Orchestration**: Docker.
  - **Cloud Platforms**: AWS (S3, EC2,...), Azure (Azure APIM, Azure Function,...).
  - **Messaging Systems**: RabbitMQ.
  - **Authentication, Authorization**: Experience of SSO using Keycloak, Auth0 and JWT.
  - **CMS**: Basic knowledge of Strapi and Moodle.
- **Databases**:
  - **SQL**: MySQL, PostgreSQL, SQL Server.
  - **NoSQL**: MongoDB.
  - **Caching**: Redis.

•Non-Technical Skills

(Other Skills)

- **Project Management Methodologies**: Experienced in both Agile/Scrum and Waterfall methodologies; adaptable to various team workflows and delivery models.
- **Language**: Fluent in reading, writing English documents and intermediate English Communication.
- **Soft skill**: Time management, priority management and team work.

Work experiences

•Fullstack NodeJS At Nashtech *3/2024 - Now*

(Graduated from the Rookie to Engineering program - Batch 7 and transitioned to working full-time afterward)

- **Open University Moodle**: A Learning Management System (LMS) for university students and lecturers. Lecturers can create courses, quizzes, and assessments, while students can submit assignments and participate in class forums. The system is built on the LAMP stack and customized from Moodle open-source to meet Open University (OU) requirements. Developed and maintained Moodle plugins using PHP and JavaScript (e.g., OUStudio, Sketch, Assignment). Implemented unit tests (PHPUnit) and Behat tests for quality assurance. Integrated AI-powered Moodle plugins using Ollama AI to summary quiz and exam to user/teacher, and improved performance and reduced token usage by implementing Redis caching.

- **Technical Stack**: PHP, Moodle, PHPUnit, Behat test, Ollama AI
- **Asset Management**: An internal dashboard for company staff to manage users, assets, and the complete asset lifecycle. Administrators can assign assets to employees and track provisioning, usage, and retirement. Responsible for developing backend API services, as well as managing CI/CD pipelines and deployment on Azure.
- **Technical Stack**: Front End (NextJS, NextAuth, JWT), Back End (NestJS, Prisma, GraphQL)
- **Tools**: Azure Devops, Github action, AWS S3, Redis, Azure container registry, Azure Web App, Keycloak
- **Role**: Fullstack Developer

Projects

•Microleap - Food detector project *6/2025 - 12/2025*

(Freelancer project)

- **Microleap** is an AI-focused startup building applications powered by state-of-the-art large language models such as **Gemini 2.5 Pro** and **GPT-4o**. I collaborated closely with the team to design and develop backend APIs for an **AI-powered food detection and nutrition analysis service**. The system automatically generates users' daily nutritional targets (calories, carbohydrates, protein, etc.) based on personal data such as height and weight. When users upload meal images, the service leverages **computer vision and generative AI** to detect food items, estimate nutritional values from images, and return personalized reports along with dietary recommendations.
  The solution utilizes **Google Vertex AI** for AI inference and **Qdrant** as a vector database for embedding storage and similarity search. The backend architecture is built using **Azure Functions** with **C# / .NET**, secured via **Azure Entra ID (SSO)**, and managed through **Azure API Management (APIM)**. Infrastructure is provisioned and maintained using **Terraform**, with **GitHub Actions CI/CD** pipelines automatically deploying Azure Functions based on **OpenAPI specifications**.
- **Technical Stack**: Azure API Management, Azure Functions, Azure Entra ID, Terraform, GitHub Actions, C# / .NET, Google Vertex AI, Qdrant
- **Role**: Backend Developer – responsible for API design, AI service integration, cloud infrastructure, security, and CI/CD automation

Languages

•Vietnamese: Native proficiency.

•English: Fluent reading and writing; basic spoken communication.

Achievements

•Gemini Certified Educator: View Certificate

•GitHub Certificate: View Certificate

•ASP.NET Core Developer: View Certificate

•Azure AI Essentials: View Certificate